

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Medium

Question

$x = 8a(b + 9)$

The given equation relates the positive numbers a , b , and x . Which equation correctly expresses a in terms of b and x ?

Answer

- A. $a = \frac{x}{8} - (b + 9)$
- B. $a = \frac{x}{8(b+9)}$
- C. $a = \frac{8(b+9)}{x}$
- D. $a = 8x(b + 9)$

Correct Answer: B

Rationale

Choice B is correct. To express a in terms of b and x , the given equation can be rewritten such that a is isolated on one side of the equation. Since it's given that b is a positive number, $b + 9$ is not equal to zero. Therefore, dividing both sides of the given equation by $8(b + 9)$ yields the equivalent equation $\frac{x}{8(b+9)} = a$, or $a = \frac{x}{8(b+9)}$.

Choice A is incorrect. This equation is equivalent to $x = 8(a + (b + 9))$.

Choice C is incorrect. This equation is equivalent to $x = \frac{8(b+9)}{a}$.

Choice D is incorrect. This equation is equivalent to $x = \frac{a}{8(b+9)}$.